BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: BN-IX-XXIII-I

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Version-A)

Time Allowed: 20 Min | **Maximum Marks:** 12 | **Paper Code:** BN-IX-XXIII-I

Note: There are THREE sections in this paper i.e. A, B & C. Attempt section A and return it to the superintendent within the given time. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Which of the following is base physical quantity?

(A) Area
(B) Density
(C) Length
(D) Volume

Q2. The speed of car is 5m/s on straight road, its velocity is:

(A) 0m/s
(B) 5m/s
(C) 10m/s
(D) 25m/s

Q3. The force which compels a body to move in a circular path is:

(A) Gravitational
(B) Friction
(C) Centripetal
(D) Nuclear

Q4. The centripetal force on an object of mass 0.5kg moving with 4m/s in a circle of radius 2m is:

(A) 4N
(B) 5N
(C) 6N
(D) 8N

Q5. The horizontal component of force is 19N and vertical component is 11N. The resultant force makes and angle with x.axis is:

(A) 0°
(B) 30°
(C) 45°
(D) 60°

Q6. Which condition is true for balancing a see-saw?

(A) τ Clockwise = τ anti clock wise
(B) τ Clockwise < τ anti clock wise
(C) τ Clockwise > τ anti clock wise
(D) τ Clockwise ≤ τ anti clock wise

Q7. Gravitational field strength is similar to:

(A) Acceleration due to gravity
(B) Mass of body
(C) Weight of body
(D) Gravitational constant

Q8. A man pulls a trolley through 25m applying a force 400N, the work done by the man is:

(A) 4000J
(B) 8000J
(C) 10,000J
(D) 20,000J

Q9. Mathematically, gravitational potential energy can be expressed as:

(A) wgh
(B) mgh
(C) mgs
(D) mg/h

Q10. We do not feel atmospheric pressure because the pressure inside and outside of our body is:

(A) Equal
(B) Different
(C) Zero
(D) Variable

Q11. How many times specific heat of water is more than specific heat of soil?

(A) 2
(B) 3
(C) 4
(D) 5

Q12. When a solid is heated its atoms?

(A) Move forward only
(B) Move backward only
(C) Vibrate violently
(D) Remain stationary

SECTION – B

Time Allowed: 2:40Hrs | **Maximum Marks:** 32 | **Paper Code:** BN-IX-XXIII-I

> Note: Attempt section B & C accordingly.

Q2. (Marks 32) Q.No 2. Attempt any EIGHT parts of the following. All parts carry equal marks.

i. Define the following branches of physics?

a. Mechanics

b. Optics

c. Electricity & Magnetism

d. Quantum Physics

ii. Define vectors, using suitable scale graphically represent 50m displacement towards South.

iii. A 10 kg body is moving with acceleration of $4m/s^2$. What is the change in momentum in 20 seconds?

iv. Why is the posted speed for a turn lower than the speed limit on most highways?

v. By giving example explain four steps of head to tail rule to add forces?

vi. What is the second condition of equilibrium? Give its mathematical form.

vii. By giving examples explain that why fossil fuels are called non-renewable source of energy.

viii. A man takes 80s in lifting a load of 250N through a height of 10m. Find his power.

ix. Differentiate between stress and strain.

x. Define evaporation. How air pressure on the surface of liquid effects the rate of evaporation?

xi. Explain how and why camels have adapted to allow them to walk more easily in desert condition?

SECTION – C

Maximum Marks: 21 | **Paper Code:** BN-IX-XXIII-I

> Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) For uniformly accelerated motion derive 1st equation of motion.

b) A bullet accelerates the length of the barrel of a gun 1.25m long with a magnitude of $4.25\times10^5 m/s^2$. With what speed does the bullet exit the barrel?

Q4.

a) Derive an expression for orbital velocity of a satellite.

b) If a satellite orbits the earth at 2,000km above sea level, how fast must the orbiting satellite travel to maintain a circular orbit?

Q5.

a) The hydraulic lift works on the principle of Pascal, explain.

b) The small piston of a hydraulic lift has an area of $0.20m^2$. A car weighing 1.20×10^4N sits on a rack mounted on the large piston has an area of $0.90m^2$. How large force must be applied to small piston to support the car?

Q6.

a) Explain latent specific heat of fusion and latent specific heat of vaporization.

b) What is specific heat of metal substance if 135kJ of heat is needed to raise 4.1kg of the metal from 18°C to 37.2°C?